THOKALA SHILPITHA CHOWDARY

Baltimore, USA | sthokal1@jh.edu | +1 (443) 844-6466 | LinkedIn | Portfolio

EDUCATION

Johns Hopkins University Baltimore, USA
Robotics MSE (Expected Completion – 2027) Fall 2025
Coursework: Machine Learning: Deep Learning (A+), Computer Vision (A-), Deep Reinforcement Learning

Amrita Vishwa Vidyapeetham Coimbatore, India
Bachelor of Technology, Automation and Robotics Engineering, CGPA: 9.02/10 2021 – 2025
Coursework: Robot Kinematics and Dynamics, Robotics and Control, Sensors and Signal Processing, Microcontrollers and Embedded Systems, Industrial Process Automation, Mobile Robotics, Industrial Internet of Things.

EXPERIENCE

The International Institute of Information Technology (IIIT) Hyderabad, India
Research Intern – Autonomous Unmanned Aerial Vehicles (UAV) Jan 2025 – July 2025

- Developed vision-based obstacle detection system for GPS-denied UAV flight by collecting 2,500+ annotated images, training CNN models, and deploying to embedded hardware, enabling autonomous navigation in indoor environments.
- Implemented camera-based visual odometry algorithms for indoor localization, iteratively testing and refining approaches like ORBSLAM to achieve reliable position estimation under challenging real-world conditions through 10+ flight tests.
- Designed MPC trajectory controller for PX4-based UAV using Python and C++, integrating with ROS2 pipeline for waypoint navigation and automated testing in Gazebo simulation and used Docker, Linux, and CMake for reproducible development workflows.
- Maintained structured datasets, experiment logs, and deployment documentation to support repeatable evaluation and continuous model improvement. Validated detection models through sim-to-real testing in Gazebo and hardware flights, iterating to improve real-world performance by addressing lighting variation and motion blur.

Indian Institute of Technology (IIT) Palakkad, India
Research Intern – Autonomous Mobile Robot (AMR) Jun 2024 – Aug 2024

- Benchmarked classical (A*, RRT, APF) and RL-based path planning algorithms in ROS2 for AMR navigation, with RL approach achieving highest success rate in cluttered environments.
- Achieved 97.8% pathfinding success rate across 500+ trials in environments with dynamic obstacles and narrow corridors using tuned RL policy.
- Integrated LiDAR and IMU sensors on TurtleBot3 hardware, calibrating sensor fusion for accurate localization and collision avoidance.

PROJECTS

Neural Deblurring for Robust Image Stitching [Link] Aug 2025 – Dec 2025

- Built panorama stitching pipeline combining neural deblurring (fine-tuned Restormer) with SIFT and SuperPoint feature matching to handle motion-blurred images, testing on custom dataset of 200+ image pairs.
- Improved feature matching inlier ratio and reduced reprojection error by 34% through fine-tuning Restormer on custom motion-blur dataset versus SIFT-only baseline.

Underwater Vehicle Manipulator System [UVMS] Aug 2024 – May 2025

- Modeled full dynamic coupling of underwater vehicle-manipulator system in Simulink, incorporating environmental disturbances including water currents and hydrodynamic forces.
- Designed Sliding Mode Controller for UVMS trajectory tracking, validating robustness to disturbances through simulation-based performance evaluation.

Self – Balancing Omni wheel RC bike Aug 2023 – May 2024

- Reached final round of E-Yantra national competition (IIT Bombay) by implementing LQR controller for self-balancing omni-wheel robot with <3° tilt tolerance, validated through CoppeliaSim simulation and hardware testing.

Wrist Rehabilitation Robot Jan 2024 – Jan 2025

- Designed 2-DOF wrist rehabilitation robot with game-based therapy interface, fabricating 3D-printed mechanical components and integrating servo control system for adjustable resistance based on patient range of motion.

LEADERSHIP & ACTIVITIES

- Course Assistant for Computer Vision, Johns Hopkins University (Spring 2026).
- Led “ROS2 workshop” for 70+ students in collaboration with the campus robotics club.

- Publication: DISCO: Diffusion-Based Inter-Agent Swarm Collision-Free Optimization for UAVs
- Conferences: Attended ROSCon-2024 and CyPhyss conference (IISC Bangalore).

SKILLS

Programming: Python, C++, C, MATLAB.
ML/Vision: PyTorch, OpenCV, CNNs (YOLO, Faster R-CNN), SIFT/SuperPoint, ORB-SLAM.
Development Tools: Docker, Linux, CMake, Git, AutoCAD.
Robotics: ROS, ROS2, PX4, Gazebo, CoppeliaSim.